

The empirical recurrence for OEIS A253040, recovered from its tail

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Abstract

OEIS A253040 records “Empirical recurrence of order 82” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 12177$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 87$. Its last coefficient is $c_{82} = -16384 \neq 0$, so no further tail satisfies anything shorter; any order-82 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A253040 is “Number of $(n+2) \times (4+2)$ 0..3 arrays with every consecutive three elements in every row, column, diagonal and antidiagonal having exactly two distinct values, and new values 0 upwards introduced in row major order.”. It has offset 1 and begins

2638, 1866, 3119, 5860, 11904, 24232, 51292, 106308, ...

The entry states:

Empirical recurrence of order 82 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:53:02, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 12177$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 12177$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 82: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 82.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 24354$ exact terms from $n = 5$ on, it returns order 82 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{82} c_i a(n-i),$$

c_1	1	c_{22}	−640778	c_{43}	−583447519	c_{64}	352977996
c_2	−1	c_{23}	−5848378	c_{44}	2303814407	c_{65}	−310186904
c_3	7	c_{24}	−372592	c_{45}	−448656569	c_{66}	231246660
c_4	49	c_{25}	−9634476	c_{46}	1344206753	c_{67}	−160242264
c_5	−45	c_{26}	7571305	c_{47}	165287337	c_{68}	33457232
c_6	48	c_{27}	19256743	c_{48}	−2411659309	c_{69}	−8268824
c_7	−384	c_{28}	6359230	c_{49}	1192303009	c_{70}	−16943760
c_8	−1014	c_{29}	54688936	c_{50}	−3013856498	c_{71}	39971792
c_9	824	c_{30}	−44882831	c_{51}	1031084406	c_{72}	−32194304
c_{10}	−908	c_{31}	−27118023	c_{52}	719234624	c_{73}	14388864
c_{11}	9144	c_{32}	−69155177	c_{53}	−648606302	c_{74}	−11295808
c_{12}	11164	c_{33}	−189859885	c_{54}	3126544935	c_{75}	3193792
c_{13}	−6316	c_{34}	154585836	c_{55}	−1726011609	c_{76}	5345664
c_{14}	8279	c_{35}	−53126270	c_{56}	1125125678	c_{77}	−1239168
c_{15}	−125005	c_{36}	377189063	c_{57}	−659747828	c_{78}	−998656
c_{16}	−66519	c_{37}	381278065	c_{58}	−1362490212	c_{79}	101888
c_{17}	−22225	c_{38}	−271299276	c_{59}	1049452264	c_{80}	139776
c_{18}	−8192	c_{39}	325754906	c_{60}	−1203143878	c_{81}	16384
c_{19}	1070150	c_{40}	−1202917411	c_{61}	917684434	c_{82}	−16384
c_{20}	188471	c_{41}	−290014361	c_{62}	−105063194		
c_{21}	947569	c_{42}	−42097927	c_{63}	−73230700		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 87$, and it is the recurrence OEIS A253040 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{82} - c_1 x^{82-1} - \dots - c_{82}$. Its constant term is $-c_{82} = 16384$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 82 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 82, so $\deg q = 82$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 82, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 25 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 82, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -16384 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-82 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A253040.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.